

Multilingual LLM Playbook

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1. Multilingual LLM Playbook

Welcome to the **Multilingual LLM Playbook** — a practical guide for building language model applications in multilingual and multicultural settings, especially in **low-resource**, **underrepresented**, or **complex sociolinguistic** environments. This playbook serves as a bridge, transforming complex technical systems and processes into understandable and actionable knowledge.

1.1 Who is this for?

This playbook is designed for:

- Researchers in NLP, HCI, and Responsible AI
- Engineers and developers building LLM-powered products (chatbots, copilots, etc.)
- Practitioners working with **non-English** or **low-resource** languages
- Anyone interested in the **intersection of language, technology, and inclusion**

1.2 What's inside?

This guide provides structured decision-making frameworks across three main strategies:

- [Translation-based](#): Translate inputs to English, process, then translate back.
- [Off-the-Shelf Prompting](#): Use multilingual-capable LLMs with carefully crafted prompts.
- [Fine-Tuning](#): Adapt open-source LLMs to your task, domain, and culture.

It also includes guidance on:

- [Evaluation methodologies](#)
- [Interactive flowchart](#) to guide your decision
- Ethical and contextual considerations

1.3 How to use this playbook

Start with the [interactive flowchart](#) or dive directly into the sections based on your project's current stage. Each section includes:

- Key decision points
- Pros and cons
- Implementation examples
- Trade-offs and pitfalls

| This playbook is evolving! Contributions and feedback are welcome.

2. Playbook

2.1 Introduction

2.1.1 Why This Playbook?

In **multilingual and multicultural environments** – especially those with limited data or infrastructure – developers face complex trade-offs when designing LLM-based applications (chatbots, coding copilots, etc.). Major LLMs today are highly English-centric (for example, only about 7% of GPT-3's training tokens were non-English)¹, leaving many of the world's 7,000+ languages under-served. Choosing the right approach for a new language or region is rarely straightforward: it requires balancing performance, cost, and cultural relevance. Without guidance, one risks deploying an AI that **performs poorly for certain languages** (often those with scarce training data)² or that misinterprets cultural context. This playbook offers a **structured decision-making framework** to navigate these choices. Whether you are an ML engineer, NLP researcher, or app-builder, goal of this playbook is to help you make **informed, resource-aware, and context-sensitive** design decisions. The framework is grounded in current research and real-world deployments, providing actionable criteria, clear trade-offs, and practical guidance for building multilingual/multicultural LLM systems.

2.1.2 Common Pitfalls

A common development pitfall is relying on **trial-and-error** experimentation without clear criteria. Teams often test various methods (e.g. translating inputs vs. direct prompting) in an ad hoc way, hoping something will work. This can lead to wasted effort and *missed alternatives*. Crucially, an approach that succeeds for one language or task may fail for another due to underlying model biases and data gaps. For instance, neither always translating everything into English nor always using the source language is guaranteed to yield optimal results across the board. Without a principled strategy, developers may stick with suboptimal settings or overlook methods that better handle specific languages, dialects, or domains. In short, a lack of systematic guidance can cause inconsistent results and **reinforce the digital divides** we aim to bridge. This playbook addresses these pitfalls by laying out clear strategies and evaluation measures, so you can avoid guesswork and proceed with confidence.

2.1.3 Three Core Strategies

We organize the playbook around **three principal strategies** that have emerged in both research and practice:

1. **Translate to English** – *Bridge through a dominant language*. In this approach, you convert non-English input into English, process it using an English-centric LLM, and then translate the output back to the target language. This leverages the robustness of models trained on huge English corpora. Many practitioners adopt this as a quick fix because it taps into the model's strongest language. However, translation can introduce **information loss and distortion**. Nuances like idioms or culturally specific references might be lost in translation, impacting the fidelity of the response. Moreover, this approach assumes high-quality machine translation is available for the language pair; if not, errors compound. It also may not be **uniformly effective across languages or tasks** – especially for content requiring region-specific knowledge or cultural sensitivity. We will discuss when this strategy makes sense (e.g. rapid prototyping for well-supported languages) and when its limitations (like cultural misalignment) become critical, and what are the various criteria to consider while using this approach.
2. **Off-the-Shelf Prompting** – *Leverage multilingual capability as-is*. Here, you use a general-purpose LLM (e.g. GPT-4 or an open multilingual model) directly in the input's language, possibly with careful prompt design to elicit the desired output. The appeal is that state-of-the-art models **can handle dozens of languages** out of the box, thanks to training on multilingual data. This strategy avoids an external translation step, preserving the original phrasing and potentially retaining more cultural nuance in the prompt. **Prompt engineering** can help guide the model's behavior in a target language. However, performance with off-the-shelf prompting is highly variable. LLMs tend to excel in languages present abundantly in their training data and struggle in low-resource languages. In fact, even advanced models have shown *significantly weaker results* for under-represented languages; for example, prompting GPT-3.5 in several African languages (especially those with non-Latin scripts) yielded poor accuracy on a basic task. This highlights that relying on a model's built-in multilingualism can fail for languages it wasn't adequately trained on. We will outline best practices for this strategy – such as prompt phrasing, examples, or instructing the model to respond in the desired language – and discuss when it's sufficient versus when you might need more tailored solutions.
3. **Fine-Tune a Small Language Model** – *Customize a lightweight model for your needs*. This approach involves taking an open-source LLM (typically a smaller model that is feasible to retrain) and adapting it on domain-specific **multilingual data**. By fine-tuning on relevant texts (e.g. a customer support dataset in Arabic and Hindi, or legal documents in multiple languages), you imbue the model with **domain knowledge and local language proficiency** beyond what general LLMs offer. The fine-tuned model can be deployed on your own infrastructure, which is beneficial if data privacy or latency is a concern. Crucially, this strategy can ensure **cultural fidelity**: the model learns terminology and styles from the actual language context of your application, rather than relying on an English proxy. The trade-offs are non-trivial. Fine-tuning requires obtaining sufficient high-quality data in each target language and domain, which is often challenging for low-resource languages. It also demands computational resources and expertise, though techniques like parameter-efficient fine-tuning (e.g. LoRA) can mitigate this by updating only small portions of the model's weights. Indeed, fine-tuning is **resource-intensive but achievable** with careful methods, and it often yields superior accuracy on specialized, multilingual tasks. Key challenges include managing data imbalance (models might over-fit to high-resource languages if not careful) and evaluating performance across languages (since standard benchmarks are usually English-centric). In this playbook, we will guide you through when fine-tuning is worth the effort – for instance, when you need **precise control, domain expertise, or support for a language that big models underserve** – and how to do it efficiently given limited infrastructure.

Each of these strategies comes with distinct **pros and cons**. The optimal choice depends on your context: the language(s) you need to support, the availability of data and computing resources, and the importance of preserving cultural nuances. In the chapters that follow, we will dive into each approach in depth, providing criteria to help you decide which strategy (or combination of strategies) suits your scenario. We then discuss **evaluation methods** for multilingual systems – how to reliably compare performance across languages and ensure fairness – and conclude with practical guidance on **deploying culturally aware** multilingual LLM systems. By approaching the design space methodically, you can move beyond guesswork and build applications that are both effective *and* inclusive in our increasingly global AI landscape.

-
1. Mondshine, I., Paz-Argaman, T., & Tsarfaty, R. (2025). Beyond english: The impact of prompt translation strategies across languages and tasks in multilingual llms. arXiv preprint arXiv:2502.09331. ←
 2. Kazemi, S., Gerhardt, G., Katz, J., Kuria, C. I., Pan, E., & Prabhakar, U. (2024). Cultural fidelity in large-language models: an evaluation of online language resources as a driver of model performance in value representation. arXiv preprint arXiv:2410.10489. ←

2.2 Evaluation Strategies for Large Language Models

- 1. Introduction: The Imperative of Robust LLM Evaluation
- 2. Core Methodologies for LLM Evaluation
 - Human-Centric Evaluation: The Gold Standard and Its Practicalities
 - Pairwise Comparison (e.g., Elo Ratings and Battle Generation)
 - Direct Assessment (Metric-based Scoring)
 - Ethical Considerations in Human Annotation and Data Collection
 - Guidelines
 - LLM-as-a-Judge: A Scalable Paradigm
 - Prompting Strategies for LLM Evaluators
 - The Critical Need for Calibration with Human Judgments
 - Identifying and Mitigating Biases in LLM-as-a-Judge
- 3. Strategic Dataset Discovery and Creation
 - Navigating Existing Benchmarks: Strengths, Limitations, and Coverage
 - Systematic Creation of New Evaluation Datasets
 - Principles of Prompt Curation for Diverse Content and Quality
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 - Ensuring Annotation Quality: Inter-Annotator Agreement Metrics
- 4. Addressing Critical Challenges in LLM Evaluation
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 - Methods for Detection in Commercial Models (Black-Box Testing)
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 - Importance of Culturally-Nuanced and Independently Created Benchmarks
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- Conclusions and Recommendations

2.2.1 1. Introduction: The Imperative of Robust LLM Evaluation

Large Language Models (LLMs) have demonstrated remarkable capabilities, yet their true performance, especially across diverse linguistic and cultural landscapes, remains opaque. Robust and comprehensive evaluation is crucial for selecting appropriate LLM approaches and validating their efficacy in global applications.

The LLM evaluation landscape is rapidly evolving, facing challenges beyond traditional benchmarks. Rapid LLM development saturates existing NLP benchmarks, making them ineffective for discerning genuine advancements. Pervasive data contamination, where test datasets inadvertently influence model training, further complicates accurate capability measurement. This necessitates dynamic, adaptive evaluation frameworks tailored for multilingual and multicultural contexts.

Standard benchmarking has inherent limitations. Many public benchmarks are absorbed into LLM pre-training data, leading to test dataset contamination where models recall memorized information rather than demonstrating true generalization. This demands dynamic benchmarking strategies to circumvent data leakage.

Traditional automated metrics like ROUGE and BLEU are often inadequate for nuanced LLM outputs. They rely on exact word matches and overemphasize length, failing to capture subjective quality, coverage, or coherence crucial for human perception of language. Their reference-based nature also requires costly human-generated gold standards that may not align with human judgments of quality. Evaluating LLM text generation is particularly challenging due to the subjective criteria of linguistic fluency, factual accuracy, and contextual appropriateness. The continuous evolution of LLMs, with training data incorporating public benchmarks, creates a perpetually shifting evaluation target. This mandates a shift from static evaluations to continuous, adaptive, and dynamic benchmarking strategies, potentially integrating human oversight or mechanisms for generating novel evaluation instances.

2.2.2.2. Core Methodologies for LLM Evaluation

Assessing LLMs requires a multifaceted approach, combining the irreplaceable insights of human judgment with the scalability offered by automated systems. This section delineates the primary methodologies employed in LLM evaluation, highlighting their respective strengths, practical considerations, and inherent complexities, with a particular focus on their application in multilingual and multicultural settings.

Human-Centric Evaluation: The Gold Standard and Its Practicalities

Human evaluation is the gold standard for assessing LLM quality and performance. It uniquely captures subtle nuances of language, context, and subjective quality often missed by automated metrics, particularly in the realm of complex, open-ended text generation tasks and diverse cultural expressions.

PAIRWISE COMPARISON (E.G., ELO RATINGS AND BATTLE GENERATION)

This method involves presenting human annotators with two distinct responses generated by different models (or different versions of the same model) for an identical prompt, asking them to select the superior response or indicate a tie. This comparative approach directly assesses relative performance.

Elo Ratings, adapted from chess, rank models based on pairwise comparison outcomes, providing a robust relative performance measure. The process involves generating comparisons, including duplicates with flipped response orders to verify annotator consistency and detect positional biases. For example, the PARIKSHA study conducted 90,000 human evaluations across 10 Indic languages using a pairwise comparison setting, inspired by LMSys ChatbotArena, to compare 30 models systematically.

For how to compute Elo ratings, you can refer to [LMSYS Orgs' implementation for Chatbot Arena Leaderboard](#). You can also find a archived copy of the notebook [here](#).

For humans to carry out such pairwise comparisons it is important that the guidelines are clear. A sample instruction from Pariksha is shown below for reference.

[Task instructions provided to the annotators for pair wise comparisons.](#)

DIRECT ASSESSMENT (METRIC-BASED SCORING)

Direct assessment involves human annotators rating a single query-response pair against predefined metrics. This provides granular understanding of a model's performance across quality dimensions.

Common direct assessment metrics include:

- **Linguistic Acceptability (LA):** Evaluates if text sounds natural to a native speaker, checking for mechanical translation or non-idiomatic expressions.
- **Task Quality (TQ):** Measures adherence to prompt instructions and incorporation of key input information.
- **Hallucination (H):** Assesses factual grounding in input and consistency with general knowledge, identifying fabricated or counterfactual claims.
- **Output Content Quality (OCQ):** Evaluates overall content standard, checking for repetition, non-native elements, or scraped text.
- **Problematic Content (PC):** Identifies offensive, inappropriate, or harmful content.

The PARIKSHA study, for instance, involved direct assessment of 8,640 data points by human annotators scoring responses on LA, TQ, and H using a comprehensive rubric across multiple languages, and the following image shows the instructions given to the annotators.

Task instructions provided to the annotators for direct assessments.

ETHICAL CONSIDERATIONS IN HUMAN ANNOTATION AND DATA COLLECTION

Human evaluation is resource-intensive, requiring significant financial investment and time. Ethical considerations are paramount, especially with diverse cultural groups.

- **Fair Compensation:** Annotators must receive fair compensation, often above local minimum wage, ensuring dignified digital labor.
- **Annotator Demographics:** Annotators must be native speakers of the languages they assess. Understanding the annotator pool is crucial for identifying and mitigating biases in human judgments, particularly in multicultural contexts.
- **Training and Guidelines:** Rigorous training and clear, consistent, detailed guidelines are essential, often refined through pilot studies to ensure cultural sensitivity in interpretation.
- **Annotator Safety:** For tasks involving toxic or sensitive content, human annotators must be protected. LLM evaluators can perform preliminary safety assessments in such cases.

Achieving high-quality, ethical, and scalable human-centric evaluation is inherently challenging. Human evaluation is precise but costly and time-consuming. Ethical considerations like fair compensation and annotator safety further increase complexity. This tension necessitates hybrid evaluation systems and LLM-as-a-judge approaches, which must be rigorously validated against human performance and adhere to strict ethical guidelines to be reliable.

GUIDELINES

LLM-as-a-Judge: A Scalable Paradigm

LLMs capable of evaluating other LLMs' outputs offer significant advancements in evaluation. This approach provides enhanced scalability and cost reductions compared to human evaluation, making it appealing for broad multilingual assessments. LLM evaluators can quantify subjective criteria like coherence, relevance, tone, and helpfulness, which traditional statistical metrics often cannot.

PROMPTING STRATEGIES FOR LLM EVALUATORS

The effectiveness of LLM evaluators is highly dependent on prompt design, which can significantly alter performance across languages and cultures.

- **Zero-shot vs. Few-shot:** While few-shot examples commonly enhance general LLM tasks, studies on LLM-as-a-judge suggest they may not substantially improve evaluator performance or human agreement. This differs from general industry assertions on few-shot learning's bias mitigation.
- **Single vs. Compound Calls:** Evaluating a single metric per LLM call ("single call") generally yields superior results and higher human concordance than evaluating multiple metrics in one "compound call." This accuracy comes at the cost of increased API calls.
- **Simple vs. Detailed Instructions:** Highly detailed, rubric-like instructions can paradoxically slightly reduce percentage agreement (PA) with human judgments. However, this may lead to a less skewed score distribution, though it doesn't eliminate the bias towards high scores. Evaluation prompts are often in English, as native language instructions can sometimes diminish performance for evaluators.

Following code snippet demonstrates usage of [LangChain's OpenEvals](#) for judging hallucinations using OpenAI' models

```

1  from openevals.llm import create_llm_as_judge
2
3  # same as 'from openevals.prompts import HALLUCINATION_PROMPT'
4  HALLUCINATION_PROMPT = """You are an expert data labeler evaluating model outputs for hallucinations. Your task is to assign a score based on the
5  following rubric:
6
7  <Rubric>
8      A response without hallucinations:
9      - Contains only verifiable facts that are directly supported by the input context
10     - Makes no unsupported claims or assumptions
11     - Does not add speculative or imagined details
12     - Maintains perfect accuracy in dates, numbers, and specific details
13     - Appropriately indicates uncertainty when information is incomplete
14 </Rubric>
15
16 <Instructions>
17     - Read the input context thoroughly
18     - Identify all claims made in the output
19     - Cross-reference each claim with the input context
20     - Note any unsupported or contradictory information
21     - Consider the severity and quantity of hallucinations
22 </Instructions>
23
24 <Reminder>
25     Focus solely on factual accuracy and support from the input context. Do not consider style, grammar, or presentation in scoring. A shorter, factual
26 response should score higher than a longer response with unsupported claims.
27 </Reminder>
28
29 Use the following context to help you evaluate for hallucinations in the output:
30
31 <context>
32 {context}
33 </context>
34
35 <input>
36 {inputs}
37 </input>
38
39 <output>
40 {outputs}
41 </output>
42
43 If available, you may also use the reference outputs below to help you identify hallucinations in the response:
44
45 <reference_outputs>
46 {reference_outputs}
47 </reference_outputs>
48 """
49
50
51 inputs = "What is a doodad?"
52 outputs = "I know the answer. A doodad is a kitten."
53 context = """
54     A doodad is a self-replicating swarm of nanobots. \
55     They are extremely dangerous and should be avoided at all costs. \
56     Some safety precautions when working with them include wearing gloves and a mask.
57     """
58
59 llm_as_judge = create_llm_as_judge(
60     prompt=HALLUCINATION_PROMPT,
61     feedback_key="hallucination",
62     model="openai:o3-mini",
63 )
64
65 eval_result = llm_as_judge(
66     inputs=inputs,
67     outputs=outputs,
68     context=context,
69     reference_outputs="",
70 )
71
72 print(eval_result)
73
74 ...
75 {
76     'key': 'hallucination',
77     'score': False,
78     'comment': '...'
79 }
80 ...

```

THE CRITICAL NEED FOR CALIBRATION WITH HUMAN JUDGMENTS

Despite their scalability, LLM evaluators' reliability hinges on rigorous calibration against human judgments. This is crucial in multilingual settings and for low-resource languages, where linguistic and cultural nuances significantly impact accuracy. LLM judgments can be inconsistent and susceptible to various influences. Calibration involves comparing LLM scores with aggregated human scores using metrics like Percentage Agreement (PA), Fleiss' Kappa (κ), and Kendall's Tau (τ) to ensure LLM evaluations reflect human perception across languages.

IDENTIFYING AND MITIGATING BIASES IN LLM-AS-A-JUDGE

LLM evaluators are prone to systemic biases that compromise assessment integrity, especially in diverse linguistic and cultural contexts:

- **Positive Score Bias (Over-optimistic Nature):** LLMs often assign higher scores in direct assessment than humans. They may fail to detect hallucinations accurately and assign high LA and TQ scores even when human annotators deem quality unsatisfactory. This bias is more pronounced in non-Latin script and low-resource languages.
- **Self-Bias:** LLMs can favor their own outputs or those from their architectural family. GPT-based evaluators, for instance, consistently rank GPT's outputs more favorably.
- **Verbosity Bias:** Both human and LLM evaluators may favor longer responses, particularly for moderate length differences (40-100 words). This bias typically diminishes with excessively long responses, which may contain irrelevant content.
- **Position Bias:** Response order can influence LLM judgments. However, some studies report low position bias when options are flipped.
- **Decisiveness (Fewer Ties):** LLM evaluators are more definitive, choosing fewer "tie" outcomes than humans in pairwise comparisons. They are also more likely to select a response even if both options contain hallucinations (e.g., 87% for LLMs vs. 53% for humans in one study).
- **Cultural Nuance Bias:** LLM evaluators show lower agreement with human judgments on culturally nuanced responses. This suggests insufficient cultural context, particularly evident in direct assessment for languages like Bengali and Odia.

The biases of LLM-as-a-judge, such as overly positive scoring and reduced hallucination detection, can create a misleading perception of superior performance. An LLM might confidently assign high scores, implying strong quality, even if content is questionable or human evaluators disagree. This is exacerbated in multilingual and culturally nuanced contexts where LLMs may lack deep understanding. The inconsistency of few-shot learning's utility in LLM evaluation further indicates that common LLM optimization strategies may not apply to nuanced evaluation tasks. This risk of misrepresentation necessitates stringent and continuous calibration against human judgments, particularly in critical applications, ensuring scalability does not compromise reliability.

2.2.3 3. Strategic Dataset Discovery and Creation

Effective LLM evaluation relies on high-quality, representative datasets. This section explores leveraging existing benchmarks and systematically creating new ones, emphasizing considerations for multilingual and multicultural contexts.

Navigating Existing Benchmarks: Strengths, Limitations, and Coverage

Initial LLM evaluation typically involves surveying and using existing benchmarks. These provide a convenient starting point with structured tasks and predefined metrics. However, their utility, especially in complex multilingual evaluation, faces significant constraints.[1, 3]

Existing benchmarks offer a broad foundation, covering diverse NLP tasks and languages, providing a common ground for model comparison.

Despite their utility, existing benchmarks have notable limitations:

- **Contamination:** Many popular benchmarks are inadvertently or intentionally included in LLM training data, making them unsuitable for fair assessment of true generalization capabilities, as performance may reflect memorization.
- **Linguistic Diversity and Resource Levels:** There's a significant imbalance in linguistic representation; Indo-European and Latin-script languages are overrepresented, while benchmarks for low-resource or non-Latin script languages are scarce, posing a challenge for comprehensive multilingual evaluation.[1, 1, 1, 1, 3]
- **Cultural Nuance:** Many multilingual benchmarks are direct translations of English originals, leading to loss of critical linguistic and cultural context. [1, 4] This can reduce LLM evaluator agreement with human judgments on culturally sensitive evaluations and perpetuate biases, making them less reliable for multicultural assessment.[4]
- **Task Coverage:** There's a deficiency in benchmarks designed to simulate real-world LLM usage, especially for complex tasks like abstract reasoning, conversational dialogue, and nuanced chat interactions across diverse languages.
- **Difficulty Level:** Older benchmarks are often too simplistic for current state-of-the-art models, leading to "ceiling effects" where models achieve near-perfect scores, hindering meaningful performance differentiation for multilingual model selection.[3]

The following table summarizes key multilingual LLM evaluation benchmarks, detailing their task types, language coverage, and notable characteristics or challenges. This is essential for navigating the current landscape and identifying appropriate datasets for specific global evaluation needs.

Benchmark Name	Task Type(s)	Language Coverage	Key Characteristics/Challenges
XNLI	Natural Language Inference	15 languages	Translated
IndicXNLI	Natural Language Inference	11 Indic languages	Culturally-nuanced, Indic languages
GLUECOS NLI	Natural Language Inference	2 languages (English-Hindi)	Code-mixed
PAWS-X	Paraphrase Identification	7 languages	High contamination risk
XCOPA	Commonsense Reasoning	10 languages	High contamination risk
XStoryCloze	Commonsense Reasoning	11 languages	Low contamination risk
TyDiQA-GoldP	Question Answering	9 languages	High contamination risk
MLQA	Question Answering	6 languages	High contamination risk
XQUAD	Question Answering	11 languages	High contamination risk
IndicQA	Question Answering	10 Indic languages	Indic languages
AfriQA	Question Answering	10 African languages	African languages, no context passage
MaRVL	Visual Question Answering	5 languages	Multimodal, culturally-driven concepts
UDPOS	Part of Speech Tagging	38 languages	Low contamination risk
PAN-X	Name Entity Recognition	48 languages	High contamination risk
XRISAWOZ	Task-Oriented Dialogue	6 languages	Dialogue, code-mixed, high contamination risk
WinoMT	Responsible AI (Gender Bias)	8 languages	Gender bias assessment
GLUECOS Sentiment Analysis (En-Es-CS)	Sentiment Analysis	2 languages (English-Spanish)	Code-mixed
Jigsaw	Toxicity Classification	6 languages	Low contamination risk
XLSum	Summarization	44 languages	Low contamination risk
IN22	Machine Translation	14 (evaluated) / 22 (total) Indic languages	Translation benchmark, general & conversational domains
XM-3600	Image Captioning	20 (evaluated) / 36 (total) languages	Multimodal, geographically diverse images
Belebele	Multiple Choice Reading Comprehension	23 (evaluated) / 122 (total) languages	Parallel, discriminates comprehension levels

Table 1: Overview of Key Multilingual LLM Evaluation Benchmarks

Systematic Creation of New Evaluation Datasets

When existing benchmarks are inadequate due to contamination, insufficient language coverage, or task irrelevance, systematically creating new, high-quality evaluation datasets becomes essential. This requires meticulous planning to ensure the dataset accurately reflects desired evaluation criteria across diverse linguistic and cultural contexts.

PRINCIPLES OF PROMPT CURATION FOR DIVERSE CONTENT AND QUALITY

Robust evaluation datasets are built on carefully curated prompts that capture the richness of human language and culture.

- **Native Speaker Involvement:** Active participation of native speakers is paramount for capturing intricate linguistic and cultural nuances in diverse regions. Prompts should be developed independently for each language, following consistent guidelines, rather than direct translation. This ensures authentic integration of local and cultural contexts, critical for meaningful multilingual assessment.
- **Content Categories:** If evaluation is domain agnostic then prompts should cover diverse domains (e.g., health, finance, culturally-specific topics) for comprehensive evaluation of LLM capabilities across knowledge areas and cultural sensitivities.
- **Systematic Generation of Varying Quality:** For meta-evaluation (evaluating LLM evaluators), a balanced dataset with "good-quality" and "bad-quality" samples is crucial. This can be achieved by systematically prompting LLMs (e.g., GPT-4) with varied temperature settings (e.g., low temperature for high quality, high temperature for lower quality) and adversarial instructions to elicit specific undesirable outputs.

DESIGNING HUMAN ANNOTATION PROTOCOLS AND GUIDELINES

The integrity of a new dataset relies heavily on well-defined human annotation protocols and clear, unambiguous guidelines that account for linguistic and cultural specificities.

- **Clear Task Definition:** Annotators must precisely understand the evaluation task and the metrics to be assessed.
- **Detailed Metric Rubrics:** Comprehensive rubrics for each metric (e.g., LA, TQ, H, OCQ, PC) are essential. These should include clear scoring ranges and illustrative examples, ideally in the target language, to ensure consistent interpretation across annotators and cultural contexts.
- **Evaluation Settings:** Implementing both pairwise comparison (comparing two responses) and direct assessment (rating a single response) captures different facets of LLM quality and performance across languages.
- **Blinding:** Annotators should be blinded to model names to prevent bias.
- **Justification:** Requiring annotators to provide justifications (e.g., audio recordings) offers valuable qualitative data for understanding their reasoning, especially for nuanced multilingual outputs.
- **Consistency Checks:** Duplicate pairings with flipped response orders in pairwise evaluations verify annotator consistency and detect positional biases.
- **Gibberish Check:** For direct assessment, an initial "gibberish" check streamlines annotation; nonsensical outputs are automatically assigned the lowest score, reducing detailed evaluation needs.

ENSURING ANNOTATION QUALITY: INTER-ANNOTATOR AGREEMENT METRICS

To confirm the reliability and quality of human annotations, rigorous measurement of inter-annotator agreement (IAA) is indispensable, particularly when aggregating judgments from diverse linguistic backgrounds.

- **Percentage Agreement (PA):** A straightforward raw measure of agreement proportion, though it doesn't account for chance agreement.
- **Fleiss' Kappa (κ):** A more robust measure factoring out chance agreement, useful for per-datapoint agreement among multiple annotators, crucial for multilingual datasets. Kappa values > 0.6 generally indicate substantial agreement.
- **Kendall's Tau (τ):** A non-parametric statistic for assessing correlation between rankings, useful for comparing human and LLM evaluator leaderboards.

Creating evaluation datasets, especially for subjective judgments, involves constructing a reliable approximation of human judgment. This relies on meticulous prompt curation, detailed annotation guidelines, and rigorous inter-annotator agreement checks. The primary goal is to minimize noise and bias in the human-labeled "gold standard," as this standard calibrates LLM evaluators. If the human-derived "gold truth" is compromised, any calibrated LLM evaluator will inherit those flaws, undermining the entire evaluation, particularly in complex multilingual scenarios.

2.2.4 4. Addressing Critical Challenges in LLM Evaluation

LLM evaluation faces significant challenges beyond methodological design, including pervasive data contamination and the complexities of multilingual and multicultural assessment, amplified by LLMs' global reach.

The Pervasive Issue of Test Data Contamination

Test data contamination occurs when test datasets, or portions of them, are inadvertently included in LLM training or fine-tuning data. This skews evaluation results, hindering accurate assessment of true multilingual capabilities.

IMPLICATIONS FOR MODEL PERFORMANCE AND TRUSTWORTHINESS

Contamination has several critical implications:

- **Inflated Performance:** It artificially inflates perceived model capabilities. Models may appear to perform exceptionally by recalling memorized answers, rather than genuinely understanding and generating responses, which is particularly misleading for claims of cross-lingual generalization.
- **Misleading Benchmarks:** Contamination exacerbates benchmark saturation. Models may seem to achieve or surpass human performance by recall, undermining benchmarks as true progress indicators, especially for multilingual advancements.
- **Hindrance to Research:** Contamination makes it challenging to discern genuine architectural improvements or training methodology advancements. Observed performance gains may be erroneously attributed to innovation when they are artifacts of data leakage.
- **Widening Digital Divide:** For non-English languages, where LLM development and evaluation are less explored, contamination can obscure actual performance gaps, hindering effective multilingual model development and potentially exacerbating the digital divide by misrepresenting capabilities in underserved languages.

METHODS FOR DETECTION IN COMMERCIAL MODELS (BLACK-BOX TESTING)

Detecting contamination in commercial LLMs is challenging due to proprietary training data. Black-box testing methods are employed:

- **Perturbation-based Method:** Golchin and Surdeanu (2023a) describe prompting the target model to generate three "perturbations" of existing test data points. These, plus the original text, are presented as four options, and the model selects its preference. Contamination is quantified using Cohen's Kappa (κ), a chance-adjusted accuracy metric. To account for positional bias, κ is adjusted to κ_{fixed} . Studies on GPT-4 and PaLM2 revealed high contamination for most datasets (e.g., PAWS-X, TyDiQA, XNLI, XCOPA), with GPT-4 generally showing higher rates, indicating widespread contamination across multilingual benchmarks.
- **Handling Long Contexts with Toolkits:** For long contexts in Question Answering (QA) tasks, especially for low-resource languages where tokenizers may over-tokenize text, the `LangChain` library is used for retrieval strategies. This involves indexing context chunks with embeddings (e.g., `text-embedding-ada-002`) and retrieving the closest chunk to the question to fit the model's context size. While the methodology is described, direct code snippets for `LangChain`'s multilingual application for this purpose are not available in the research material, though `LangChain` generally offers 'How-to Guides' for RAG use cases.²

METHODS FOR DETECTION IN OPEN-SOURCE MODELS

For open-source models, the "Black Box test" (Oren et al., 2023) offers a statistical method to provide provable guarantees of contamination. This test leverages "exchangeability," where example order can be shuffled without altering joint distribution.

- **Canonical vs. Shuffled Order Preference:** If a model was exposed to a benchmark during training, it will show a statistically significant preference for the "canonical order" (original sequence in public repositories) over randomly shuffled orderings. If this difference is significant, the dataset is contaminated for that model. Empirical tests on instruction-tuned Llama2, Mistral, and Gemma 7B variants indicated contamination in datasets like PAWS-X, XCOPA, XQUAD, and XRISAWOZ, highlighting the widespread nature of this issue even in open-source multilingual models.

Data contamination fundamentally transforms LLM generalization measurement into memorization reflection. This leaves LLM capabilities a "known unknown," especially for tasks with exceptionally high reported performance. The widespread nature of contamination across commercial and open-source models points to a systemic issue in the LLM development ecosystem. The emphasis on high benchmark scores often overlooks evaluation integrity. This challenges trust in reported scores and necessitates dynamic, contamination-aware evaluation strategies, coupled with greater transparency from model developers on training data composition, particularly for multilingual datasets.

Nuances of Multilingual and Multicultural Evaluation

Evaluating LLMs in non-English languages presents a complex array of linguistic, cultural, and technical challenges, demanding specialized approaches for accurate assessment.

THE IMPACT OF TOKENIZER FERTILITY ON PERFORMANCE AND COST

Tokenizer fertility (average sub-words per tokenized word) critically influences pre-trained multilingual model performance, directly impacting efficiency and cost.

- **Higher Fertility, Worse Quality/Higher Cost:** Tokenizers (e.g., OpenAI's) are less efficient for low-resource, non-Latin script languages (e.g., Malayalam, Tamil), resulting in very high fertility rates (approx. 10 sub-words per word). This inefficiency leads to higher costs, as more tokens are needed for input encoding and response generation via API calls, creating an economic barrier for multilingual applications.
- **Correlation with Performance:** A statistically significant negative correlation exists between tokenizer fertility and dataset-specific performance. Models perform worse on languages where their tokenizers are less efficient, highlighting a fundamental technical challenge in multilingual LLM development.

IMPORTANCE OF CULTURALLY-NUANCED AND INDEPENDENTLY CREATED BENCHMARKS

Many existing multilingual benchmarks are direct translations of English originals, losing crucial linguistic and cultural context.[1, 4] This can lead to lower LLM evaluator agreement with human judgments on culturally nuanced responses. Evaluation prompts should be developed independently by native speakers for each target language, following consistent guidelines, rather than being mere translations. This ensures accurate capture of local and cultural nuances in evaluation material, leading to more authentic and reliable multicultural assessments.

The combination of inefficient tokenizers, limited pre-training data for low-resource languages, and reliance on translated benchmarks creates a "cultural blind spot" in global LLMs. Even grammatically correct text in a low-resource language may lack the deep cultural context for truly nuanced, appropriate, and helpful responses. This deficiency is evident in subjective tasks or direct assessment where cultural understanding is paramount. This "cultural blind spot" is not just a performance limitation but an ethical concern, potentially exacerbating the "digital divide" by making models less useful or even harmful to diverse populations. This necessitates a deliberate focus on culturally-aware AI development and evaluation practices.

2.2.5 5. Best Practices and Future Directions

Rigorous LLM evaluation is an evolving discipline requiring continuous adaptation. This section outlines best practices for robust evaluation frameworks and identifies promising future research avenues, particularly for multilingual and multicultural LLM development.

Recommendations for Designing and Implementing Robust Evaluation Frameworks

To effectively navigate LLM evaluation complexities, several key practices should be adopted:

- **Dynamic and Adaptive Benchmarking:** Transition from static, easily contaminated benchmarks to dynamic systems generating novel evaluation instances. This is crucial for countering data contamination and keeping pace with LLM advancements, especially across diverse languages.
- **Culturally-Appropriate Design:** Prioritize developing culturally-nuanced benchmarks created independently by native speakers, not machine translations, to ensure authenticity and relevance for specific cultural contexts.
- **Multi-Dimensional Evaluation:** Employ diverse metrics (LA, TQ, H, OCQ, PC) to comprehensively capture LLM performance facets, including subjective qualities and cultural appropriateness.[1, 2]
- **Transparent Reporting (Bender Rule):** Model releases should explicitly state training and evaluation languages. Clear, empirically-backed language support statements foster transparency and set appropriate user expectations for multilingual models.
- **Confidence Intervals:** Always include and report confidence intervals in evaluation results. This enhances replicability and facilitates more reliable inferences about model performance differences across languages and tasks.
- **Consistent and Replicable Implementations:** Meticulously document, ensure consistency, and reproduce all evaluation setups, including prompting strategies, hyperparameters, and data processing steps. This is vital for fair comparison of multilingual LLMs.
- **Focus on Appropriate Difficulty:** Select or develop benchmarks offering sufficient challenge for state-of-the-art models. This avoids ceiling effects and provides meaningful signals for multilingual model selection and improvement.

The Role of Hybrid Human-LLM Evaluation Systems

Given the complementary strengths of human judgment (for nuance and gold-standard quality) and LLM-as-a-judge capabilities (for scalability), a hybrid evaluation system is often the most robust and practical solution for comprehensive multilingual assessment.

- **Human-in-the-Loop Calibration:** All LLM-based multilingual evaluations must be rigorously calibrated against human-labeled judgments for each language before deployment. This is essential for identifying and mitigating biases like positive score bias and cultural nuance bias, ensuring automated evaluation reliability.
- **Strategic Allocation of Resources:** LLMs can efficiently handle large-scale preliminary evaluations, filtering and categorizing outputs. Human experts can then focus their valuable time on assessing complex, ambiguous cases, evaluating cultural nuances, and performing final validation, particularly for low-resource languages where LLM performance may be less reliable.
- **Iterative Refinement:** LLMs can generate initial evaluations or iteratively refine their own outputs. In this symbiotic relationship, human feedback serves as the ultimate arbiter, guiding and validating automated assessments, leading to more refined multilingual models.

Continuous Adaptation of Benchmarks and Methodologies

LLM evaluation is dynamic, requiring ongoing research to keep pace with rapid model advancements, especially in expanding multilingual and multicultural coverage.

- **Advancing Prompting Strategies:** Continuous research is needed to develop more sophisticated prompting approaches for LLM evaluators, including automatic prompt tuning adaptable for various languages.
- **Exploring Diverse Evaluator Models:** Future work should investigate the efficacy of smaller LLMs or models trained with broader non-English data coverage for evaluation tasks, aiming to reduce reliance on English-centric evaluators.
- **Balanced Dataset Creation:** There's a need for more balanced calibration datasets, ensuring diverse human judgment distribution across quality levels and languages.
- **Developing Evaluator Personas:** Exploring various evaluator personas within LLMs could represent diverse human perspectives and facilitate consensus-building in automated evaluations, reflecting multicultural viewpoints.
- **Expanding Evaluation Dimensions:** Future research should broaden evaluation to include fairness, bias, robustness, and efficiency, particularly for non-English languages where dedicated datasets for these aspects are currently scarce.

2.2.6 6. Evaluation Software and Frameworks

The LLM evaluation landscape is supported by a growing ecosystem of software, frameworks, and toolkits that streamline and standardize assessment. These tools facilitate benchmarking, metric calculation, and human-in-the-loop calibration. While specific code snippets for their direct implementation in multilingual or multicultural contexts are typically found in their documentation rather than in the provided research material, the following tools are notable for enabling robust evaluation strategies:

General LLM Evaluation Frameworks and Harnesses:

- **EleutherAI LLM Evaluation Harness**⁴: A widely used tool for evaluating large language models.
- **OpenAI Evals**⁶: An evaluation tool provided by OpenAI.

LLM-as-a-Judge Tools:

- **LLM Comparator from PAIR Google**¹⁴: A side-by-side evaluation tool that facilitates human-driven LLM evaluation.
- **OpenEvals from LangChain**¹⁵: An evaluation framework that supports LLM-as-a-judge methodologies.
- **Confident-AI DeepEval**¹⁷: An LLM Evaluation Framework offering unittest-like evaluation of LLM outputs, often leveraging LLM-as-a-judge.

2.2.7 Conclusions and Recommendations

The analysis of LLM evaluation reveals a complex landscape with significant challenges. Evaluation is a central, indispensable component in the LLM lifecycle, especially for models intended for global, multilingual, and multicultural use. Traditional static benchmarks are insufficient due to pervasive test data contamination, which inflates reported performance and obscures true generalization. This contamination, found in both commercial and open-source models, creates a "known unknown" regarding LLM competence, undermining trust in published benchmarks.

Furthermore, the evaluation of LLMs in multilingual and multicultural contexts presents profound difficulties. A consistent performance gap exists between English and non-English languages, particularly for low-resource languages and non-Latin scripts. This is exacerbated by inefficient tokenizers, which increase costs and negatively correlate with performance, and by reliance on translated benchmarks lacking cultural nuance. This creates a "cultural blind spot," where LLMs may produce grammatically correct text but fail to provide culturally appropriate or deeply contextualized responses, potentially widening the digital divide.

While LLM-as-a-judge offers scalable evaluation, it has biases, including overly positive scoring, self-bias, and verbosity bias, creating an "illusion of competence" when human judgments diverge. The inconsistent effectiveness of common LLM prompting strategies, like few-shot learning, highlights the unique demands of evaluation tasks. The continuous evolution of LLMs means evaluation must be dynamic, reflecting a "moving target."

Based on these findings, the following recommendations are put forth for designing and implementing robust LLM evaluation frameworks, with a strong emphasis on multilingual and multicultural considerations:

1. **Prioritize Dynamic and Adaptive Benchmarking:** Move away from static, contaminated benchmarks towards systems generating novel evaluation instances. This ensures evaluations measure true generalization for diverse linguistic inputs.
2. **Invest in Culturally-Nuanced Dataset Creation:** Develop new evaluation datasets with native speaker involvement, curating prompts independently for each language to capture local and cultural subtleties, rather than relying on direct translations.
3. **Implement Hybrid Evaluation Systems:** Combine human judgment's quality and nuance with LLM-as-a-judge's scalability. LLMs handle large-scale preliminary assessments, while human experts focus on complex cases, cultural validation, and final arbitration, especially for low-resource languages.
4. **Mandate Rigorous Calibration of LLM Evaluators:** All LLM-based multilingual evaluations must be rigorously calibrated against human-labeled judgments for each language. This is crucial for identifying and mitigating biases like positive score bias and cultural insensitivity.
5. **Enhance Transparency and Reproducibility:** Model developers should explicitly declare training and evaluation languages, providing evidence-backed language support statements. Evaluation reports should consistently include confidence intervals and detail all methodological aspects for replicability across linguistic contexts.
6. **Address Tokenizer Inefficiencies:** Further research and development are needed to improve tokenizer efficiency for low-resource and non-Latin script languages, reducing costs and improving performance in these critical areas.
7. **Broaden Evaluation Scope:** Expand evaluation beyond traditional accuracy metrics to encompass critical dimensions like fairness, bias, robustness, and efficiency, particularly for non-English languages where dedicated datasets for these aspects are currently limited.

LLM evaluation is an ongoing, iterative development cycle. The future success of LLMs, particularly in serving a global, diverse user base, hinges on continuously refining evaluation methodologies, fostering an ecosystem as dynamic and sophisticated as the models it assesses, with a deep understanding of multilingual and multicultural nuances.

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1. <https://github.com/guidance-ai/guidance/tree/main> ↵
 2. <https://github.com/hwchase17/langchain> ↵
 3. https://github.com/anoopkunchukuttan/indic_nlp_library ↵
 4. <https://github.com/EleutherAI/lm-evaluation-harness> ↵
 5. <https://github.com/microsoft/eureka> ↵
 6. <https://github.com/openai/evals> ↵
 7. ((<https://github.com/microsoft/YourBench>)) ↵
 8. (<https://www.google.com/search?q=https://github.com/LLMeBench/LLMeBench>) ↵
 9. <https://github.com/bigcode-project/bigcode-evaluation-harness> ↵
 10. <https://github.com/allenai/zeroeval> ↵
 11. <https://github.com/embeddings-benchmark/mteb> ↵
 12. <https://github.com/OpenICL/OpenICL> ↵
 13. <https://github.com/epfl-lm/lm-pub-quiz> ↵
 14. <https://github.com/google/lm-comparator> ↵
 15. <https://github.com/langchain-ai/langchain/tree/master/libs/langchain/langchain/evaluation> ↵
 16. <https://github.com/mozilla/lm-buddy> ↵
 17. <https://github.com/confident-ai/deepeval> ↵
 18. <https://www.arize.com/phoenix> ↵
 19. <https://www.mlflow.org/docs/latest/llms/lm-evaluate/index.html> ↵
 20. <https://github.com/langfuse/langfuse> ↵
 21. <https://github.com/truera/trulens> ↵
 22. <https://github.com/explodinggradients/ragas> ↵
 23. (<https://www.google.com/search?q=https://github.com/NVIDIA/garac>) ↵
 24. <https://github.com/microsoft/autogenbench> ↵
 25. <https://github.com/microsoft/copilot-arena> ↵
 26. (<https://www.google.com/search?q=https://github.com/NVIDIA/score>) ↵
 27. <https://github.com/microsoft/prompty> ↵
 28. <https://github.com/promptfoo/promptfoo> ↵
 29. <https://github.com/Chainlit/chain-forge> ↵
 30. <https://github.com/ironclad/rivet> ↵

2.3 Translation

2.3.1 Overview

One common strategy for multilingual deployment is to use **automatic translation**:

- Translate non-English input to English
- Use an English-centric LLM
- Translate the output back to the original language

This approach leverages the strength of English-dominant LLMs without requiring native multilingual support.

2.3.2 When to Use

- Your language has **reliable machine translation** to/from English
- You lack resources to fine-tune models
- **Accuracy** is more important than **cultural fluency**

2.3.3 Pros

- Works with **off-the-shelf translators** and English LLMs
- Fast to set up
- Useful when **native-language LLM support is weak**

2.3.4 Cons

- **Cultural nuance** may be lost
- Translation errors can **propagate** to final output
- Limited **personalization** (e.g., forms of address, dialects)

2.3.5 Translation Strategies

Full Translation

Translate entire input and output to/from English.

Selective Translation

Translate only parts of the prompt:

- Keep examples or instructions in original language
- Translate only the query or expected output

This modular approach helps balance **fidelity** and **model compatibility**.

Scratchpad Techniques

Use translation as an internal reasoning tool:

- Translate input to English, reason in English, then translate back
- Effective in **very low-resource** languages

Fine-Tuning Translators

Train or adapt translation systems:

- Generate **synthetic data** to expand training
- Use **instruction fine-tuning (IFT)** with LoRA
- Post-process outputs to improve fluency and tone

2.3.6 Risks & Caveats

- **Dependency on translation quality**
- Loss of **idiomatic** or **culturally embedded** meaning
- Introduces **two points of failure** (input and output)

2.3.7 Summary

Translation is a powerful strategy — especially when paired with robust English LLMs — but requires careful design and evaluation to preserve meaning, tone, and cultural context.

2.4 Off-the-Shelf Prompting

2.4.1 Overview

This strategy involves using **pretrained general-purpose LLMs** (e.g., GPT-4, Claude, Phi, Llama) without additional training — relying instead on smart **prompt engineering** to handle multilingual or multicultural requirements.

2.4.2 When to Use

- You need a **fast prototype** or **proof-of-concept**
- The target language is **reasonably supported** by the model
- You can invest in **prompt design**, **few-shot examples**, or **retrieval augmentation**

2.4.3 Pros

- **No training or labeling needed** — rapid time-to-market
- Works across many **mid- to high-resource languages**
- Easy to experiment with **cultural framing** (e.g., personas, tones)
- Effective for **zero-shot** and **few-shot** tasks

2.4.4 Cons

- **Prompting is fragile** — performance can vary widely
- **Inconsistent behavior** in low-resource languages
- Hard to control **tone and style** over long conversations
- May underperform on **sensitive or domain-specific tasks**

2.4.5 Prompting Techniques

- **Prompt Sensitivity**: test how different phrasings affect performance
- **Multilingual Prompting**:
 - Prompt in English or native language
 - Use hybrid prompts
- **Cultural Prompting**:
 - Frame answers using local context
 - e.g., "Answer as if you're a journalist from Japan"
 - Use **prompt programming frameworks** to iterate and test

2.4.6 Key Considerations

- **Language Coverage**: which languages does the LLM support reliably?
- **Latency & Cost**: does your setup require real-time responses?
- **Domain Fit**: how well does the LLM generalize to your task?

2.4.7 Summary

Prompting off-the-shelf LLMs is a fast, versatile approach for multilingual tasks — especially when infrastructure is limited. However, the effectiveness hinges on thoughtful prompt design, language support, and robust testing.

2.5 Fine-Tuning

2.5.1 Overview

Fine-tuning involves adapting a **lightweight open-source LLM** (e.g., Mistral, Phi-2, Gemma) on **domain-specific and culturally relevant multilingual data**.

This strategy is suitable for building highly customized, culturally aware, and privacy-preserving systems.

2.5.2 When to Use

- You have access to **domain-specific data**
- You need **full control** over tone, behavior, or cultural alignment
- You're deploying in **low-resource, private, or air-gapped** environments

2.5.3 Pros

- Highly **customizable and controllable** outputs
- Works well for **niche domains** and **local dialects**
- Supports **offline or private deployment**
- Enables **value alignment** through custom objectives

2.5.4 Cons

- Requires **labeled data** or curated corpora
- Demands **training infrastructure** and model expertise
- Smaller models may struggle with **complex reasoning**
- Cultural generalization is challenging with **limited data**

2.5.5 Training Techniques

Parameter-Efficient Fine-Tuning (PEFT)

- e.g., LoRA, QLoRA
- **Pros**: minimal changes to core model, efficient adaptation
- **Cons**: managing multiple adapters can complicate deployment

Full Fine-Tuning

- Complete retraining of the model for maximum alignment
- More resource-intensive, but useful when PEFT falls short

Continued Pretraining

- Language/domain adaptation via additional unlabeled text
- Can precede or complement fine-tuning

2.5.6 Data Preparation

- Collect or augment with:

- **Parallel texts, glossaries, instruction datasets**
- **Local stories, FAQs, proverbs** for cultural grounding
- Generate **synthetic data** when resources are limited
- Translate English IFT datasets
- Apply **selective translation** or **cultural augmentation**

2.5.7 Cultural Alignment

- Fine-tune with culturally diverse data
- Avoid overfitting to narrow perspectives
- Explore **self-pluralizing alignment**:
- Teach models to present **multiple cultural viewpoints**

2.5.8 Summary

Fine-tuning offers maximum control and cultural alignment, but requires careful planning around data, infrastructure, and deployment. It's ideal when your application demands high fidelity to local language, norms, or values.

2.6 Conclusion

2.6.1 Final Thoughts

Designing multilingual and culturally sensitive LLM systems is a complex challenge, particularly in settings with limited resources, fragmented datasets, and underrepresented languages.

This playbook has offered **three core strategies** — Translation, Off-the-Shelf Prompting, and Fine-Tuning — each suited to different scenarios depending on language support, data availability, technical capacity, and deployment goals.

2.6.2 Choosing the Right Path

Use this decision framework to:

- Evaluate trade-offs between speed, quality, cost, and cultural fidelity
- Choose approaches aligned with your domain and audience
- Consider evaluation, data, and infrastructure from the outset

2.6.3 Checklist

Do you have a reliable evaluation setup (benchmark or custom)?

Is translation sufficient or does your application demand cultural grounding?

Can prompting alone meet your accuracy and customization needs?

Do you have the data and compute for fine-tuning a small model?

2.6.4 Beyond the Playbook

This guide is not exhaustive. It reflects the state of practice and research as of now. The multilingual NLP ecosystem is rapidly evolving, and we expect continued improvements in:

- Open-source multilingual models
- Language-specific datasets
- Cultural alignment techniques
- Evaluation frameworks and tooling

2.6.5 Acknowledgments

Inspired by deployments, research prototypes, and community discussions across academia and industry.

We hope this playbook helps you make **strategic, culturally thoughtful, and resource-aware** choices.

If you found this helpful or would like to contribute, we welcome feedback and collaboration.

3. Playbook Flowchart

flowchart TD

```
A[Start: Define Task and Language] --> B{Does an Evaluation Dataset Exist?}
B -- Yes --> B1[Check for Contamination]
B1 --> B2[Run Evaluation Protocol]
B2 --> B3[Choose Evaluation Metrics]
B3 --> B4[Analyze Output Quality]

B -- No --> C{Create Synthetic Dataset?}
C -- Yes --> C1[Prompt-Based Generation]
C1 --> B3
C -- No --> C2[Human Annotation Process]
C2 --> C3[Set Fair Wages and Guidelines]
C3 --> B3

B4 --> D{Is Machine Translation Reliable?}
D -- Good --> D1[Use Translate-to-English Strategy]
D -- Moderate --> D2[Use Off-the-Shelf Prompting]
D -- Poor --> D3[Fine-Tune a Small Model]

D1 --> D1a[Evaluate Cultural Loss Risk]
D1a --> D1b[Try Selective Translation]
D1b --> D1c[Consider Translator Fine-Tuning]

D2 --> D2a[Design Multilingual Prompts]
D2a --> D2b[Test Prompt Sensitivity]
D2b --> D2c[Use Prompt Engineering Tools]

D3 --> D3a[Collect Domain-Specific Data]
D3a --> D3b[Choose Fine-Tuning Method]
D3b --> D3c[Generate or Translate IFT Data]
D3c --> D3d[Align with Cultural Values]

D1c --> E[Re-evaluate Model Output]
D2c --> E
D3d --> E

E --> F{Ready to Deploy?}
F -- Yes --> G[Deploy and Monitor]
F -- No --> H[Refine or Iterate Further]

G --> I[End of Process]
```